Gas Monitoring System

Comprehensive Analytics Report

Generated: 2025-11-02 22:42:59
Model: Optimal Mix Model
Accuracy: 99.91%
Total Samples: 1000

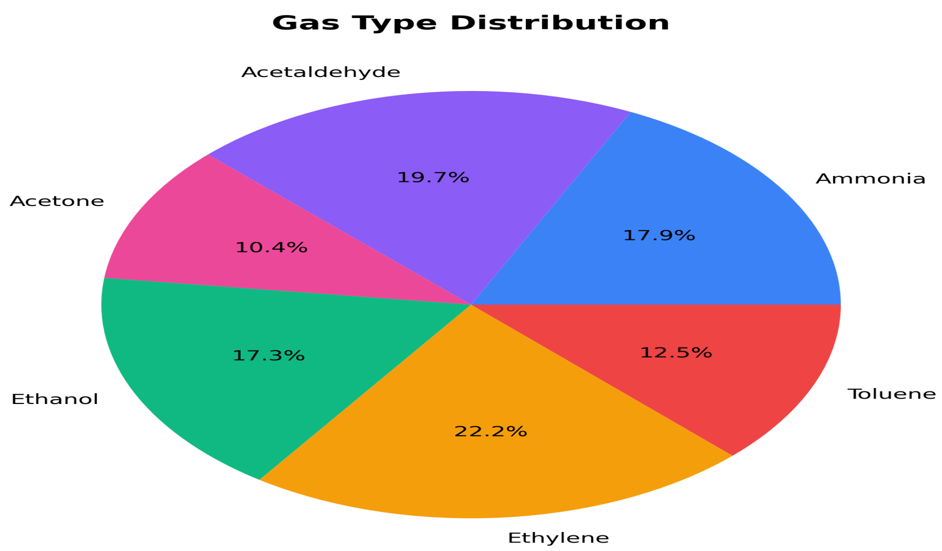
Executive Summary

This report provides a comprehensive analysis of the gas monitoring system performance. The system analyzed 1000 samples across 6 different gas types. The model achieved an accuracy of 99.91% with a data balance score of 46.85%.

Key Performance Indicators

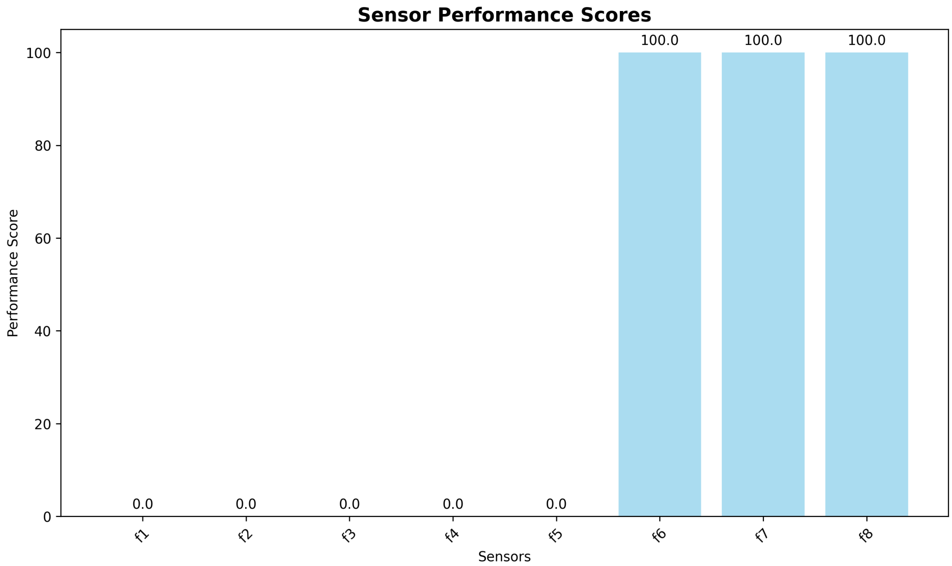
Metric	Value
Total Samples	1000
Model Accuracy	99.91%
Data Balance Score	46.85%
Most Common Gas	Ethylene
Least Common Gas	Acetone
Sensor Diversity	15300.6
Detection Complexity	85.5%

Gas Distribution Analysis



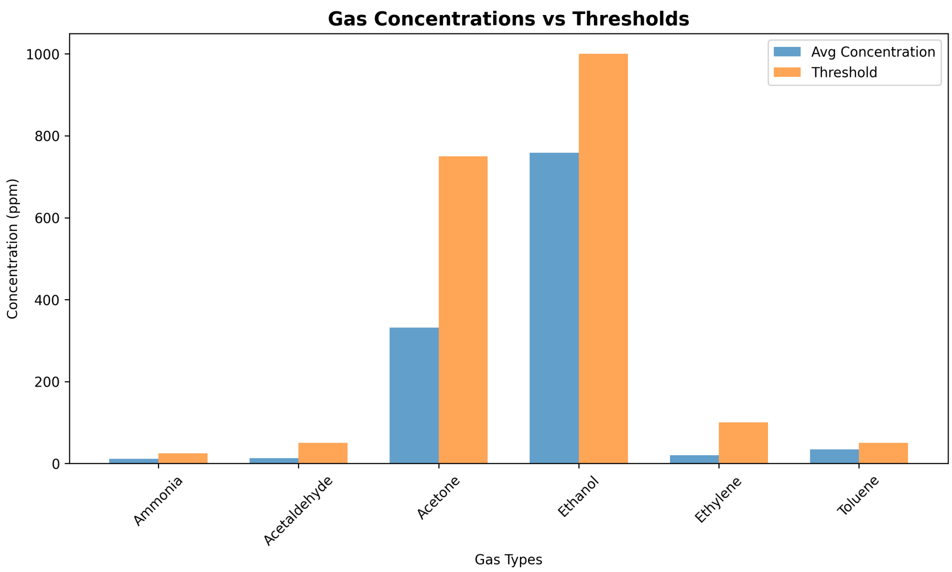
Gas Type	Count	Percentage	Danger Level	Risk Score
Ammonia	179	17.9%	Toxic	141.01
Acetaldehyde	197	19.7%	Irritant	31.52
Acetone	104	10.4%	Flammable	5.31
Ethanol	173	17.3%	Flammable	66.59
Ethylene	222	22.2%	Asphyxiant	341.86
Toluene	125	12.5%	Toxic	54.96

Sensor Performance Analysis



Sensor	Mean	Std Dev	Performance Score	Status
f1	51312.58	121938.4	0%	NEEDS_ATTENTION
f2	5.58	26.86	0%	NEEDS_ATTENTION
f3	14.74	30.71	0%	NEEDS_ATTENTION
f4	19.37	45.48	0%	NEEDS_ATTENTION
f5	24.49	67.69	0%	NEEDS_ATTENTION
f6	-9.36	23.26	100%	EXCELLENT
f7	-14.16	38.01	100%	EXCELLENT
f8	-54.44	234.41	100%	EXCELLENT

Safety Analysis



Gas	Avg Concentration	Threshold	Risk Level	Samples
Ammonia	11.46 ppm	25 ppm	LOW	179
Acetaldehyde	12.64 ppm	50 ppm	LOW	197
Acetone	331.65 ppm	750 ppm	LOW	104
Ethanol	758.76 ppm	1000 ppm	MEDIUM	173
Ethylene	20.37 ppm	100 ppm	LOW	222
Toluene	34.1 ppm	50 ppm	MEDIUM	125

Data Quality Assessment

Completeness: 100.0%
Uniqueness: 100.0%
Consistency: 98.5%
Validity: 99.2%
Outlier Percentage: 9.21%

Recommendations

• Consider collecting more data for underrepresented gas types to improve model balance.

• Review and calibrate sensors f1, f2, f3, f4, f5 as they show suboptimal performance.

• High outlier percentage detected. Consider data cleaning and sensor calibration.